

Group Work

1. At what speed a hypothetical train should move so that its length will be one-third of the stationary length?
2. The length of a rocket on the ground is $100m$. When rocket moves with respect to a stationary observer on the ground its length appears to be $99.5m$. Calculate the speed of the rocket.
3. A space shuttle crosses the football ground of BUET along of its length at speed of $0.6c$. The football ground is $100m$ long and $50m$ wide. What is the length and width of the football ground according to the measurement by a foreign astronaut in the space shuttle?
4. An astronaut at the age of 25 years went to the space by a space-shuttle moving with $1.8 \times 10^8 ms^{-1}$ velocity. According to earth he spent 30 years in space. What is his age now?
5. Show by the theory of relativity that the velocity of a body cannot be more than the velocity of light.
6. Explain the theory of relativity is not applicable for the very short velocity of particle compared to the velocity of light.
7. Write down a report on Einstein's Theory of Relativity and express your opinion in its favour and against it.
8. Established the relation between the total energy and momentum.